S-BLOCK

- last e^- enters in outermost S-orbital.

S-BLOCK

Group 1 — alkali metals

bcoz rxn with water → Hydroxide (Strongly alkaline nature)

- GEC: $[1G]\ ns^1$
- Li → 2nd Period → $[He]\ 2s^1$
  Na → 3rd " → $[Ne]\ 3s^1$
  K – 4 " → $[Ar]\ 4s^1$
  Rb – 5 " → $[Kr]\ 5s^1$
  Cs – 6 " → $[Xe]\ 6s^1$
  Fr – 7 " → $[Rn]\ 7s^1$ (Radioactive)
- oxidation state = +1
- Monovalent M^{+1} ion.

Group 2 — alkaline earth metals

bcoz oxide & hydroxide → alkaline nature & found in earth crust

Exception: Be (not alkaline earth metal)

- GEC: $[1G]\ ns^2$
- Be → $[He]\ 2s^2$
  Mg → $[Ne]\ 3s^2$
  Ca → $[Ar]\ 4s^2$
  Sr → $[Kr]\ 5s^2$
  Ba → $[Xe]\ 6s^2$
  Ra → $[Rn]\ 7s^2$
- oxidation state = +2
- Divalent M^{+2} ion.

first element of each group shows diff behavior (DIAGONAL)

Diagonal relationship: Li → Mg, Be → Al

diagonal relationship is due to same ionic size or same charge/size ratio

Atomic / Ionic Radius

Alkali

- Period: Largest element in a particular period
- Group: Down the group size ↑ (bcoz no of shell ↑ increasing down the group)

Alkaline

- Period → smaller than alkali metals bcoz Z_{eff} ↑ (from L to R)
- Group → down the group size ↑. (no of shell ↑ down the group)

Ionization enthalpy

- Period: Low in Period (Size ↑)
- Group: Size ↑ (decreases down the group)

Size ↑ → outermost e^- are weakly attracted by nucleus.

Alkali

Period: IE_1 more I.E than alkali bcoz group 2 has small size than alkali.

IE_2 → alkali has more IE_2 (bcoz of inert gas configuration)

Group: Size ↑ So, I.E ↓

Hydration Energy

$HE \propto \frac{charge}{size}$

less size ⇒ more hydration

$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ [HE]

Li has max hydration so Lithium salts are most hydrated. ex → $LiCl \cdot 2H_2O$

less size ⇒ more hydration

$Be^{+2} > Mg^{+2} > Ca^{+2} > Sr^{+2} > Ba^{+2}$

B/w alkali and alkaline :-

alkali → Large size

alkaline → Small size

∴ alkaline has more hydration Energy than alkali.

So, mostly hydrated compounds are formed by group 2 elements.

$MgCl_2 \rightarrow MgCl_2 \cdot 6H_2O$

$CaCl_2 \rightarrow CaCl_2 \cdot 6H_2O$

NaCl ⟶ ✗

KCl ⟶ ✗

Physical property

- Silverywhite, soft, light metal
- density
  - Period → Large size → density ↓
  - Group → ↑ down the group.

exception $Li < K < Na < Rb < Cs$

- MP & BP → Low MP & B.P
  bcoz weak metallic bond
  bcoz only one valence e⁻.

- Silvery white, soft, lustrous. (~~Reactivity~~ harder than alkali)
- Be & Mg → Greyish Color.
- MP & B.P → MP & B.P is more than alkali (bcoz of small size, bcoz of good metallic bo, bcoz of two valence e⁻.
- Electropositive character e⁻ ~~size more → e⁻ can easily be~~ size ↑ e⁻ loosing tendency ↑ ∴ +ve cha
- increases down the group.

  $Be < Mg < Ca < Sr < Ba$

  less tendency to loose e⁻.

Flame color

Li → crimson Red

Na → yellow

K → voilet

Rb → Red voilet

Cs → Blue

Be → ✗

Mg → ✗

Ca → Brick Red

Sr → crimson

Ba → apple green.

NOTE : Both alkali & alkaline have high electrical & thermal conductivit

Chemical Property

- High reactive (large size) & low I.E
- ~~down the~~ group :→ Size↑ : Reactivity↑

Reactivity towards air → रासायनिक क्रिया होगा

- ~~Alkali metals tarnish in dry air (formation of oxide)~~
- dry air + Alkali metal → oxide
  ↓ due to this alkali metals tarnish
- moisture + Alkali → Hydroxide

Reaction with O_2

Li → Oxide (O^{-2}) → Li_2O

Na → Peroxide (O^{-1}) → Na_2O_2

K, Rb, Cs → Superoxide ($O^{-1/2}$) → KO_2, RbO_2, CsO_2

Reaction with N_2

$Li + N_2 \rightarrow Li_3N$

Rest do not give these type of rxn.
only Li forms Nitride.

Kept in kerosene oil bcoz of high reactivity (air & water)

Li → Paraffin wax

- less reactive than alkali (size less than alkali)
- Group : Size↑ ⇒ Reactivity↑

- Be + (air & water) → No rxn
- Mg + (air & water) → No rxn

↳ inert towards oxygen & water.
bcoz oxide film is formed on their surface

Powdered form reacts with O_2 & N_2.

$Be + O_2 \rightarrow BeO$
$\quad + N_2 \rightarrow Be_3N_2$ ✓

$Mg + O_2 \rightarrow MgO$
$\quad + N_2 \rightarrow Mg_3N_2$ =

$Ca + O_2$ & $N_2 \rightarrow CaO$, Ca_3N_2

$Sr + O_2$ & $N_2 \rightarrow SrO$, Sr_3N_2

$Ba + O_2$ & $N_2 \rightarrow BaO$, Ba_3N_2

Ca, Sr, Ba + water → They React with even cold water

$Ca + H_2O$ (cold) → $Ca(OH)_2$
$Sr + H_2O$ (cold) → $Sr(OH)_2$
$Ba + H_2O$ (cold) → $Ba(OH)_2$

- $Mg + H_2O$ (cold) → ✗
  $Mg + H_2O$ (hot) → $Mg(OH)_2$
- $Be + H_2O$ (hot) → ✗
  $Be + H_2O$ (cold → ✗

Reaction with H_2O

- Metal + H_2O → Hydroxide + H_2↑

$Li + H_2O \rightarrow Li(OH) + H_2$ → Lithium reacts less vigorously with water. (bcoz of small size & high HE)
$Na + H_2O \rightarrow NaOH + H_2$
$K + H_2O \rightarrow KOH + H_2$
$Rb + H_2O \rightarrow RbOH + H_2$
$Cs + H_2O \rightarrow CsOH + H_2$

Rest, other metal reacts explosively with H_2O

Reaction with Hydrogen

Metal + H_2 → Metal hydride (MH)

Li + H_2 → LiH } → 1073K

Na + H_2 → NaH

K + H_2 → KH } 673K

Rb + H_2 → RbH

Cs + H_2 → CsH

- all these metal hydrides are ionic solids (high melting point)

metal + H_2 → Metal hydride (MH)

Be + H_2 → X

Mg + H_2 → MgH_2

Ca + H_2 → CaH_2

Sr + H_2 → SrH_2

Ba + H_2 → BaH_2

~~Another way of Preparation of BeH_2~~

$BeCl_2 + LiAlH_4$ → BeH_2 + LiCl + $AlCl_3$

BeH_2 → ~~Hydrogen~~ Bridge Bond

3 center 2e⁻ bond

Reaction with Halogen

Metal + X_2 → MX (metal halide)

Li + X_2 → LiX → Covalent nature

(SC) LA

small cation

(LP)

Polarisability ↑

anion e⁻ cloud more distort

e⁻ cloud sharing

Covalent character

LiI is most covalent

Li^+ → SC

I^- → LA

Na + X_2 → NaX

K + X_2 → KX

Rb + X_2 → RbX

Cs + X_2 → CsX

ionic nature

Metal + X_2 → MX_2 (metal halide)

Be + X_2 → BeX_2 → BeF_2*, $BeCl_2$*

Mg + X_2 → MgX_2

Ca + X_2 → CaX_2

Sr + X_2 → SrX_2

Ba + X_2 → BaX_2

$(NH_4)_2BeF_2$ $\xrightarrow{\Delta}$ BeF_2* + NH_3

Ammonium tetrafluoroberylate

BeO + C + Cl_2 $\xrightarrow{600-800K}$ $BeCl_2$* + CO

~~Hydration Energy~~ Reducing Nature

- alkali metals are Strong Reducing agent bcoz of large negative electrode potential.
- Li → Highest Reducing Nature

Sublimation En

Ionizn En

Hydrn En

Li → High H·E → high reducing nature.

→ Alkaline earth metals are strong Reducing agent

→ These are less power Reducing agent than alkali

- Reducing nature increases from Be to Ba.

→ ~~Be ... Ba~~ ... nature.

bcoz electrode potential becomes more EN down the Group

Reaction with liq NH_3

- all alkali dissolve in liq NH_3

$$M + (x+y)NH_3 \rightarrow [M(NH_3)_x]^+ + (e^-(NH_3)_y]$$

- deep blue color soln due to ammoniated e^-
- paramagnetic soln
- Standing for long hour → H_2 release & amide form.

$$M^+_{(am)} + e^- + NH_3(l) \rightarrow MNH_2 + \tfrac{1}{2} H_2\uparrow$$

Conc. Soln

→ diamagnetic

⟷ Blue Color → Bronze Color

- all alkaline dissolve in liq NH_3.
- deep blue black solution due to ammoniated e^-.

$$M + (x+y)NH_3 \rightarrow [M(NH_3)_x]^{+2} + 2[e(NH_3)_y]^-$$

Reaction with acid

metal + acid → + $H_2\uparrow$ (release)

$$M + HCl \rightarrow MCl_2 + H_2\uparrow$$

$$M + H_2SO_4 \rightarrow MSO_4 + H_2\uparrow$$

General characteristic of Compd of Alkali

- In excess of air

$Li \rightarrow Li_2O + Li_2O_2$ (Oxide & some peroxide)

$Na \rightarrow Na_2O_2 + NaO_2$ Peroxide & some Superoxide

K, Rb, Cs → Superoxide: KO_2, RbO_2, CsO_2

- Superoxide is stable with Large size cations because Large anions are stable by large Cation through lattice Energy

Gen. Characteristic of compound of Alkaline

- These also form ionic compound but less ionic than alkali
- oxide & Hydroxide of Be & mg are more covalent than ca sr Ba.

bco2 (Be & mg) → SC → Fajan's Rule.

Oxide & Hydroxide

Metal + oxygen → MO

Be + O_2 → BeO → amphoteric → covalent nature

mg + O_2 → MgO

ca + O_2 → CaO

sr + O_2 → SrO

Ba + O_2 → BaO

→ Basic nature, ionic, rock salt type str. (NaCl type str.)

Basic oxide of G2 + H_2O → Hydroxide.

MgO → $Mg(OH)_2$

CaO + H_2O → $Ca(OH)_2$

SrO → $Sr(OH)_2$

BaO → $Ba(OH)_2$

Solubility, Thermal stability, Basic character increases downgrp.

→ Hydroxides formed are white crystalline solid

Hydrolysis of oxide/peroxide/superoxide

oxide & Peroxide is Colorless

$M_2O + H_2O \rightarrow M^+ + OH^- \rightarrow$ Colorless

$M_2O_2 + H_2O \rightarrow M^+ + OH^- + H_2O_2 \rightarrow$ Colorless

superoxide is yellow / orange (Paramagnetic)

$MO_2 + H_2O \rightarrow M^+ + OH^- + H_2O_2 + O_2 \rightarrow$ orange/yellow (Paramagnetic)

$Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

Solubility, Thermal stability, Basic Character ↑

$BeO + H_2O \rightarrow Be(OH)_2$

Amphoteric

$Be(OH)_2 + OH^- \rightarrow [Be(OH)_4]^{-2}$

Beryllate ion

$Be(OH)_2 + HCl \rightarrow [Be(OH)_4]Cl_2$

Halides

MX (X = F, Cl, Br, I)

(M = Li, Na, K, Rb, Cs)

M.P & B.P Floride > chloride > Bromide > Iodide

Preparation

oxide or Hydroxide or Carbonate + HX (aq) (Hydrohalic acid) = MX (metal halide)

- all halides are soluble in H_2O

But LiF & CsI are very low soluble in H_2O

LiF → High Lattice Enthalpy

CsI → less hydration energy

- all alkaline metal halides are ionic except Beryllium halide.

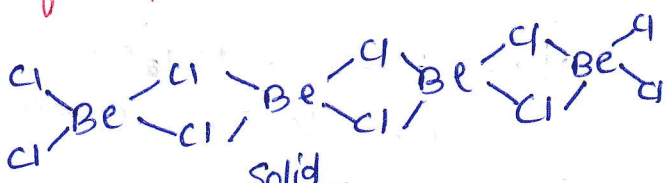
Beryliumhalide → Covalent. (Small Cation)

- $BeCl_2$

Solid state → polymer

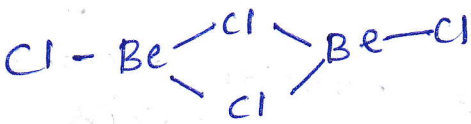
Vapourstate → dimer

if temp ≃ 1200K → monomer



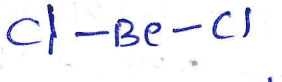
Solid





dimer

Temp – 1200K,



monomer

Tendency to form halide hydrate decreases down the group.

$MgCl_2 \cdot 8H_2O$

$CaCl_2 \cdot 6H_2O$

$SrCl_2 \cdot 6H_2O$

$BaCl_2 \cdot 2H_2O$

OXO-Acid

- What is oxoacid ??

⇒ Acidic proton (H^+) is attached to hydroxyl group. and oxo group is also attached to same ~~oooooo~~ atom O

ex. Carbonic acid (H_2CO_3) → $C(=O)(OH)_2$ or $HO-C(=O)-OH$

Sulphuric acid H_2SO_4 → $HO-S(=O)_2-OH$ or $(OH)_2S(=O)_2$

- alkali metal forms salt with oxo acid.

$M + H_2CO_3 \rightarrow M_2CO_3$ Carbonate, $MHCO_3$ bicarbonate

$M + H_2SO_4 \rightarrow$ Sulphate

$M + HNO_3 \rightarrow$ nitrate.

- Down the group, thermal stability of Carbonate & bicarbonate increases.
- Li_2CO_3 is not thermally stable.

$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

- $LiHCO_3$ → exists in liquid form, do not exist in solid form

- Carbonate : produced by passing CO_2 through their hydroxi

$M(OH)_2 + CO_2 \rightarrow M_2CO_3 + H_2O$

- Solubility of Carbon decreases down the group.

all $M_2CO_3 \xrightarrow{\Delta} MO + CO_2$

- Thermal stability ↑ down the grou

Bicarbonate $\xrightarrow{\Delta}$ carbonate $+ CO_2 + H_2O$

Sulphate

prepare:-

$M + H_2SO_4 \rightarrow MSO_4 + H_2\uparrow$

$MO + H_2SO_4 \rightarrow MSO_4 + H_2O$

$M(OH)_2 + H_2SO_4 \rightarrow MSO_4 + 2H_2O$

$MCO_3 + H_2SO_4 \rightarrow MSO_4 + CO_2 + H_2O$

- $BeSO_4$ & $MgSO_4$ } Highly soluble in water.

Be^{+2} Mg^{+2} } → greater hydration energy.

- ~~solub~~ $CaSO_4$

$SrSO_4$

$BaSO_4$ ↓ Solubility decreases from $CaSO_4$ to $BaSO_4$

nitrate

Prepare

$MO + HNO_3 \rightarrow M(NO_3)_2 + H_2O$

$M(OH)_2 + \text{"} \longrightarrow M(NO_3)_2 + 2H_2O$

$MCO_3 + \text{"} \longrightarrow M(NO_3)_2 + CO_2 + H_2O$

$MgNO_3 \cdot 6H_2O \rightarrow$ 6 water of cryst

$BaNO_3 \longrightarrow$ no water of crystalzn

all nitrate decompose to give oxide + nitrate + O_2.

$M(NO_3)_2 \xrightarrow{\Delta} MO + NO_2 + O_2$

M = Be, Mg, Ca, Sr, Ba

Anamalous property of Li

- size atomic/ionic size
- absence of d orbital
- Harder than other metals its MP & B.P are higher than other metals.
- High polarizing power (small cation) ⇓ form covalent comp
- Li forms Li_3N & Li_2O others donot.
- LiCl → deliquescent and forms $LiCl \cdot 2H_2O$ other chlorides donot form hydrate.
- $LiHCO_3$ → liq (not solid) other alkali bicarbonate → solid
- Heat $LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$

  $NaNO_3 \xrightarrow{\Delta} NaNO_2 + O_2$
- LiF, Li_2O, Li_2CO_3, Li_3PO_4 are less soluble in H_2O while others are more soluble in H_2O.

Anamalous property of Be

- Small atomic/ionic size,
- absence of d orbital.
- Harder than other & high MP & BP
- Doesn't react with water (even on boiling)
- Be forms covalent comp and other form ionic.
- Be forms volatile Be_3N_2 while others form non volatile M_3N_2.
- BO & $Be(OH)_2$ are amphoteric while others are basic oxide & hydroxide.
- Be can ~~only~~ exhibit CN not more than 4 due to absence of d orbital while other have CN = 6

Diagonal Relnship b/w Li & Mg

- Both are harder & lighter in group.
- Li & Mg ⟶ react slowly with water (
- Hydroxide & oxide are less soluble
- Hydroxide decompose on heating
- Both form nitride Li_3N, Mg_3N_2
- Li_2O, MgO } ⟶ donot give superoxide
- $LiCO_3 \xrightarrow{\Delta} Li_2O + CO_2$
  $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$
- $LiHCO_3$, $Mg(HCO_3)_2$ } is not found solid state
- LiCl & $MgCl_2$ } deliquesent
- $LiCl \cdot 2H_2O$, $MgCl_2 \cdot 8H_2O$ } both form hydrate

Relatn b/w Be & Al

- Be, Al } + Acid → no rxn
  (bcoz of Presence of oxide film on meta surface).
- $Be(OH)_2$, $Al(OH)_3$ } + excess OH^- → $[Be(OH)_4]^{2-}$, $[Al(OH)_4]^-$
- $BeCl_2$, $AlCl_3$ } vapour phase → Bridge str
  Banana Bond
  3 centre 2 e^-
- Both form Complex
  BeF_4^{-2} & AlF_6^{-3}